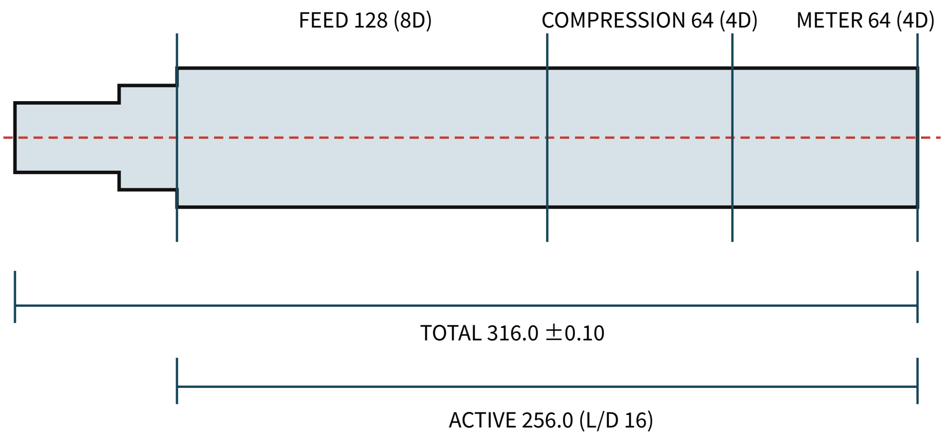
1. 16 mm × 16 L/D screw/barrel RFQ drawing

Revision: `solid-manifold-openmodelica-v0.4` / 단위: mm / 온도: 20 ±2 °C / 일반 모서리 C0.2–0.5, burr 없음

FULL 발주 HOLD — EX-CPN-SCR/EX-CPN-BAR 공정 coupon, relief coupon과 공급사 DFM, Gate-3 승인 전 EX-SCR-01/EX-BAR-01/EX-DIE-01…05 발주 금지

1.1. EX-SCR-01

EX-SCR-01 — 16 mm × 16D single screw RFQ drawing



Rear: Ø12 h6 x35, keyseat 4 P9 x2.5 deep · thrust journal Ø15 h6 x20 · neck 5

OD 15.92 -0.02/0 · pitch 16.00 ±0.03 · land 1.60 ±0.05 · single start RH

root Ø10.88 feed → linear Ø14.08 compression → Ø14.08 meter · end faces ⊥ A 0.03

Datum A: common journal axis · flight OD TIR ≤0.05/256 · drive-to-flight concentricity ≤0.03 · OD Ra≤0.8 µm

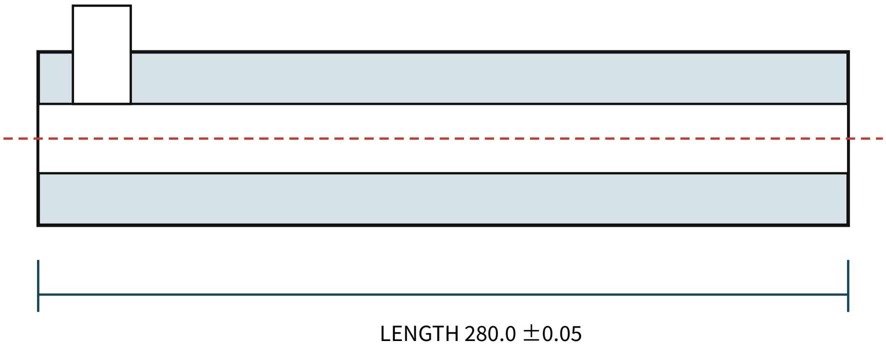
SCM440 QT 28–32 HRC → gas nitride 0.30–0.50 mm, surface 900–1100 HV · full part HOLD

재료	SCM440, normalized blank → rough turn → QT 28–32 HRC
주요 치수	Total 316.0 ±0.10; active 256.0; single-start RH; pitch 16.00 ±0.03; land 1.60 ±0.05; OD Ø15.92 −0.02/0
Zone/root	Feed 128, root Ø10.88; compression 64, linear Ø10.88→Ø14.08; meter 64, root Ø14.08
Journal	Drive Ø12 h6 ×35, KS/DIN key 4×4, keyseat 4 P9 wide ×2.5 +0.10/0 deep; thrust Ø15 h6 ×20; shoulder perpendicularity 0.03 to Datum A; flight start 0° ±5° from key plane
GD&T	Flight OD TIR ≤0.05/256; drive-to-flight concentricity ≤0.03; straightness ≤0.05/256
표면	Flight OD Ra≤0.8 µm; root/flank Ra≤1.6 µm; weld repair 금지
열처리	Gas nitride effective case 0.30–0.50, surface 900–1100 HV; journals mask; final OD grind between centres

가공 route: normalized blank → datum centre drilling → rough turn → QT → finish journals/shoulders leaving flight allowance 0.15 → 4-axis flight mill → root/flank polish → gas nitride → flight OD grind → TIR/Ra/hardness 검사.

1.2. EX-BAR-01

EX-BAR-01 — Ø34 / ID16.20 barrel RFQ drawing



OD Ø34.00 ±0.05 · bore Ø16.20 +0.02/0 after final hone · radial clearance 0.14–0.16

feed port 18 axial ×20, near edge 12.0 from rear Datum B · break edge R0.5

4x M4×0.7-6H full depth8/tap drill11, PCD26 at 45° · outer/inner ligament ≥2.0/2.9 · faces B/C ⊥ D 0.03

bore Ra 0.4–0.8 µm; SCM440 QT 28–32 HRC → gas nitride 0.30–0.50 mm, ≥900 HV

Assembly: B aligns screw active start; screw tip is 24.0 behind C. Final hone after nitride; report ID at B+20/140/260.

No weld/plating on bore · feed-port centre plane is angular datum · full part HOLD

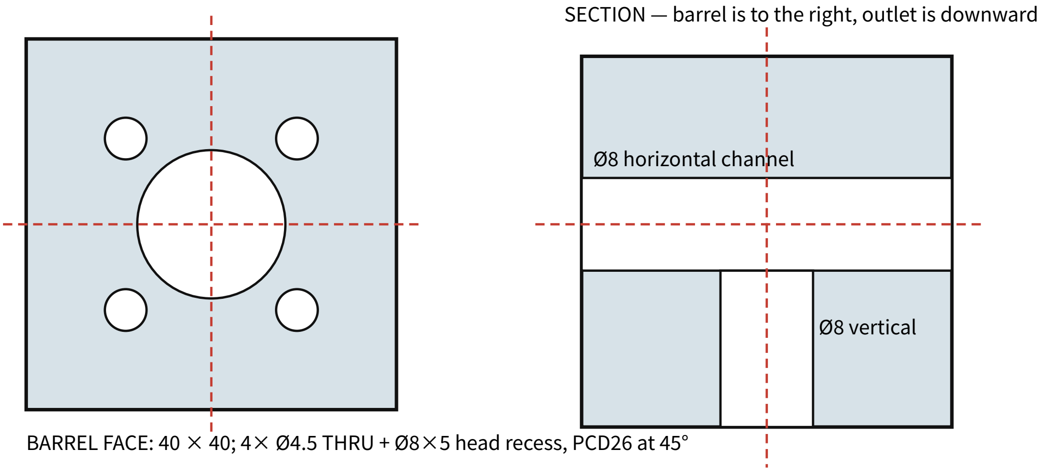
재료	SCM440 solid/seamless blank, QT 28–32 HRC
주요 치수	OD Ø34.00 ±0.05; L280.00 ±0.05; final bore Ø16.20 +0.02/0
Port/thread	18 axial ×20, rear edge 12.0 from Datum B, edge R0.5; front 4× M4×0.7-6H full depth≥8/tap drill≥11 on PCD26 at 45/135/225/315° from port centre plane; OD/bore breakthrough 금지
나사 ligament	M4 major envelope 기준 outer 2.0 mm, bore-side 2.9 mm 이상; supplier는 thread minor/major와 실제 OD/ID로 재확인
GD&T	Bore straightness ≤0.05/256; bore-to-OD/register concentricity ≤0.05; end face perpendicularity 0.03
표면	Final bore Ra 0.4–0.8 µm; no weld/plating in bore
열처리	Gas nitride 0.30–0.50, ≥900 HV; final hone 후 effective case ≥0.25
검사	ID/roundness at z=20/140/260; roundness ≤0.02; air/3-point bore-gauge record

가공 route: rough deep drill → stress relieve → semi-finish ream/hone leaving 0.05–0.08 → port/flange machine → gas nitride → final hone → ID/roundness/Ra/case-depth 검사.

Matched drawing-limit diametral clearance는 0.28–0.32, radial clearance는 0.14–0.16이다. Screw OD와 barrel ID를 세 station에서 기록하고 이 범위 안에서 pair한다.

1.3. EX-DIE-01…05 connected open-die

EX-DIE-01…05 — connected 90° open-die assembly RFQ drawing



BODY SCM440 QT 28–32 HRC + gas nitride: 40×40×48; face flatness 0.03; channels Ø8 H9;
breaker seat Ø16.20 +0.05/0 ×3; insert seat Ø12.00 +0.03/0 ×14; heater Ø6.20 H9 thru;
sensor Ø3.20 +0.05/0 blind12; 2×M4-6H depth8 retainer holes at X8/32; all melt edges R0.3.

BREAKER 304: Ø15.90 -0.05/0 ×2; 7×Ø2.00 +0.05/0 (six PCD10). INSERT 17-4PH H900:
Ø11.90 -0.02/0 ×14; Ø3.00 +0.02/0 ×10 land, Ra≤0.4; 4 mm 60° included transition; TIR≤0.02.

RELIEF 304 t1.5: 32×20; two 10×2.5 webs; 2×Ø4.5 @24; Ø4 bypass. Hot coupon 3–6 MPa.
GASKET C110 annealed t0.50: OD34 / ID16.20 / 4×Ø4.5 PCD26. Use 4×M4×45 class10.9, 3.0 N·m.

Leak/relief hot test behind grounded shield only. Analytical relief estimate is screening, not release evidence. FULL PART HOLD.

EX-DIE-01 body	SCM440 QT 28–32 HRC + gas nitride; 40×40×48; Ø8 H9 intersecting melt turn; breaker seat Ø16.20 +0.05/0 ×3; insert seat Ø12.00 +0.03/0 ×14; sealing face flatness 0.03
Mount/heating	4×Ø4.5 through + Ø8×5 head recess PCD26; heater Ø6.20 H9 through; sensor Ø3.20 +0.05/0 blind12; 2×M4-6H retainer threads depth8
EX-DIE-02 breaker	304, Ø15.90 −0.05/0 ×2; 7×Ø2.00 +0.05/0, six PCD10; flatness 0.03
EX-DIE-03 insert	17-4PH H900, OD Ø11.90 −0.02/0 ×14; outlet Ø3.00 +0.02/0 ×10 land Ra≤0.4; concentricity 0.02 to OD
EX-DIE-04 relief	304 stainless t1.5, 32×20, two 10×2.5 webs, 2×Ø4.5 at 24, Ø4 bypass; three same-lot coupons must physically open 3–6 MPa at 265 °C without insert ejection
EX-DIE-05 gasket	C110 annealed t0.50 ±0.03, OD34, ID16.20, 4×Ø4.5 PCD26; qty2 including one spare
Fastener	4×M4×45 class 10.9 at 3.0 N·m; retainer 2×M4 at 1.2 N·m; new gasket each removal

가공 route: six-face datum mill → intersecting Ø8 drill/ream → seat/bolt/heater/sensor machine → stress relieve → final seat/face → gas nitride → sealing face lap → borescope/deburr/pressure-coupon inspection. 교차 유로 step/burr는 R0.3 이하이며 weld repair와 hot-path plating은 금지한다. Relief 265 °C 계산 screening 4.32 MPa는 RFQ reference일 뿐 release evidence가 아니다.

1.4. Supplier deliverables

- Material certificate, QT hardness, nitride surface hardness/effective case depth
- Screw OD/TIR/concentricity/straightness, barrel ID/roundness/straightness report
- Ra trace, actual matched clearance table, STEP 기준 DFM deviation list
- Die intersecting-channel borescope record, seat/land/concentricity report와 same-lot relief coupon 3개 결과
- 공차 변경 제안은 발주 전 서면 승인; silent substitution 금지

1.5. Process coupon release

먼저 EX-CPN-SCR L48.00 ±0.05, three RH pitches 16.00 ±0.03, feed-zone OD/root/land와 EX-CPN-BAR L60.00 ±0.05, OD34.00 ±0.05, final ID16.20 +0.02/0만 견적·가공한다. Ends는 axis에 0.03 이내 수직이다. Coupon은 동일 material/route/heat treatment로 만들고 OD/ID, pitch, land, Ra, hardness/case depth를 검사한다. Coupon 불합격 또는 질문 미해결이면 full part는 계속 HOLD다.